Chapter 1

1. What file can you modify?

-FreeRTOSConfig.h

2. list 3 examples of hardware that could be setup

-GPIO pins
-Clock
-ADC
-DAC
-PWM

3. what are the return types

-Void and is in task.c
-Basetype_t and is in task.c
-char pointer and is in queue.c
-unsigned Basetype_t and is in semphr.h
-void pointer and is in timer.c

Chapter 2

1. How does pvPortMalloc() work differently with the different heaps

Heap_1: has a single heap stack that you can allocate tasks in.
it is deterministic with no memory fragmentation. However, there
is no vPortFree() function implemented meaning no tasks can be deleted

Heap_2: Deprecated(only exists for backwards compatibility). the start
of heap 4 but was implemented poorly.

Heap_3: Uses stdlib's malloc() and free() functions. TOTAL_HEAP_SIZE has
no effect on function calls. is thread-safe because scheduler is
temporarily suspended.

Heap_4: like Heap_1 it splits array into smaller blocks when allocating.
However, vPortFree() can be used and tasks can be deleted.

Heap_5: like Heap_4 but memory allocation can be down from multiple memory
spaces.

2. Use case for xPortGetMinimumEverFreeHeapSize(void)

-if later in the program running you need to create a new thread with Y bytes
on the stack. You could call this to see if there would be any danger creating
the task.